

ASSIGNMENT-1

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BATCH-01

TASK-1:

```
number = 5 # Change this value to compute factorial of a
different number
factorial = 1

for i in range(1, number + 1):
    factorial *= i

print(f"The factorial of {number} is {factorial}")
```

OUTPUT:

```
ISHAMADEENA@ISHAS-MacBook-Air AI-assisted code % /usr/local
The factorial of 5 is 120
```

TASK-2:

```
n = 5 # Example input
result = 1
for i in range(1, n + 1):
    result *= i
print("Factorial of", n, "is", result)

n = 5 # Example input
result = 1
for i in range(2, n + 1): # Start from 2 for clarity
    result *= i
print(f"Factorial of {n} is {result}") # Improved readability
with f-string
```

OUTPUT:

```
ISHAMADEENA@ISHAS-MacBook-Air AI-assisted code
Factorial of 5 is 120
Factorial of 5 is 120
```

TASK-3:

```
def calculate_factorial(n):  
    # Initialize result to 1 because factorial of 0 is 1  
    factorial = 1  
    # Multiply numbers from 2 to n  
    for i in range(2, n + 1):  
        factorial *= i  
    return factorial  
# Main execution block  
if __name__ == "__main__":  
    number = int(input("Enter a number: "))  
    result = calculate_factorial(number)  
    print(f"Factorial: {result}")
```

OUTPUT:

```
Enter a number: 6  
Factorial: 720  
ishamedesha@Tshas MacBook
```

TASK-4:

```
# Factorial without functions
```

```
n = int(input("Enter a number: "))  
factorial = 1  
for i in range(2, n + 1):  
    factorial *= i  
print(f"Factorial: {factorial}")
```

```
# Factorial with functions
```

```
def calculate_factorial(n):  
    # Initialize result to 1 because factorial of 0 is 1  
    factorial = 1  
    # Multiply numbers from 2 to n  
    for i in range(2, n + 1):  
        factorial *= i  
    return factorial  
  
if __name__ == "__main__":  
    number = int(input("Enter a number: "))  
    result = calculate_factorial(number)  
    print(f"Factorial: {result}")
```

OUTPUT:

```
Enter a number: 4
Factorial: 24
Enter a number: 5
Factorial: 120
```

TASK-5:

```
def factorial_iterative(n):
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result
```

```
if __name__ == "__main__":
    num = int(input("Enter a number: "))
    print("Factorial:", factorial_iterative(num))
```

OUTPUT:

```
ishamadeen@ishams-MacBook:
Enter a number: 6
Factorial: 720
ishamadeen@ishams-MacBook:
```